Trade Policy: Optimal Tariffs, Political Economy, and Trade Wars

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Roadmap

- Why would a country ever tax its own imports? Three classic motives:
  - **Terms of trade (TOT)**: a large importer pushes down the world price it pays
  - **Imperfect competition**: tariffs shift rents from foreign firms
  - **Political economy**: organized industries buy protection

1. A single welfare formula that nests all motives (Feenstra, ch. 7–9)
2. Optimal tariffs under perfect and imperfect competition; the role of pass-through
3. Protection for Sale (Grossman and Helpman, 1994) and its tests (Goldberg and Maggi, 1999)
4. Measuring market power: Feenstra (1994), Broda and Weinstein (2006), Broda, Limão and Weinstein (2008)
5. Two-country games: Nash tariffs and the purpose of the GATT/WTO (Bagwell and Staiger)
6. Quantitative trade wars and trade talks (Ossa, 2014)

Outline

1 **Welfare Effects of a Tariff**

2 Optimal Tariffs

3 Political Economy: Protection for Sale

4 Measuring Market Power in Trade

5 Nash Tariffs and Trade Agreements

6 Quantifying Trade Wars and Trade Talks: Ossa (2014)

Setup: Quasi-Linear Economy

- Household h has quasi-linear utility $c_0^h + U^h(c^h)$; the numeraire c_0 absorbs income effects
- Equally weighted social welfare, with $d^h(p)$ the demand of household h:

$$W(p, I) = \sum_{h=1}^{H} \Big[\underbrace{I^h - p\, d^h(p)}_{c_0^h} + U^h\big(d^h(p)\big) \Big]$$

- Envelope theorem: $\dfrac{\partial W}{\partial p} = -\sum_h d^h(p) \equiv -d(p)$ (Roy's identity with $\partial W / \partial I = 1$)
- Focus on **one** imported good with a specific tariff t:
  - Domestic price $p = p^* + t$, where p^* is the world (tariff-exclusive) price
  - Imports $m = d(p) - y$, domestic output y with cost $C(y)$
  - The ad valorem case is analogous

Total Effect of a Tariff

- Income = labor income + tariff revenue + domestic profits:

$$W\big[\,p,\ \underbrace{L}_{\text{labor}} + \underbrace{tm}_{\text{revenue}} + \underbrace{py - C(y)}_{\text{profit}}\,\big] \equiv W(t)$$

- Differentiate with respect to t, using $\partial W/\partial p = -d$ and $\partial W/\partial I = 1$:

$$\frac{dW}{dt} = -d\,\frac{dp}{dt} + \left(m + t\frac{dm}{dt}\right) + \left(y\frac{dp}{dt} + \big(p - C'(y)\big)\frac{dy}{dt}\right)$$

- Use $m = d - y$ to combine the dp/dt terms:

$$\frac{dW}{dt} = m\Big(1 - \frac{dp}{dt}\Big) + t\frac{dm}{dt} + \big(p - C'(y)\big)\frac{dy}{dt}$$

- Since $p = p^* + t$: $1 - \frac{dp}{dt} = -\frac{dp^*}{dt}$

Reading the Formula

Welfare effect of a tariff

$$\frac{dW}{dt} = \underbrace{t\,\frac{dm}{dt}}_{\text{deadweight loss}} \quad \underbrace{-\,m\,\frac{dp^*}{dt}}_{\text{terms of trade}} \quad + \underbrace{\big(p - C'(y)\big)\,\frac{dy}{dt}}_{\text{production efficiency}}$$

- **Deadweight loss** ≤ 0 for $t > 0$: imports shrink below the efficient level, distorting consumption and production; zero at $t = 0$
- **Terms of trade**: a gain whenever the world price falls; first order at $t = 0$, proportional to imports
- **Production efficiency**: only when $p > C'(y)$ (imperfect competition); expanding an under-producing domestic firm raises welfare

- Small country, perfect competition: $p = C'(y)$, $dp^*/dt = 0$ and $dp/dt = 1$, so $dW/dt = t\,m'(p) < 0$ for any $t > 0 \Rightarrow$ free trade is optimal

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Large Country, Perfect Competition

- Upward-sloping foreign export supply $x^*(p^*) \Rightarrow \frac{dp^*}{dt} < 0$: the domestic price rises by less than the tariff (**incomplete pass-through**)
- Perfect competition: $p = C'(y)$ and imports depend only on p, so $\frac{dm}{dt} = \frac{dm}{dp}\frac{dp}{dt}$. Set $dW/dt = 0$:

$$\frac{t^*}{p^*} = \left(\frac{dp^*}{dt}\frac{m}{p^*}\right) \Big/ \left(\frac{dm}{dp}\frac{dp}{dt}\right)$$

- **Representation 1**: market clearing $m = x^*(p^*)$ gives $\frac{dm}{dt} = \frac{dx^*}{dp^*}\frac{dp^*}{dt}$, so

$$\boxed{\frac{t^*}{p^*} = \frac{1}{\eta^*}, \qquad \eta^* \equiv \frac{dx^*}{dp^*}\frac{p^*}{x^*}}$$

- The optimal tariff equals the inverse elasticity of foreign export supply $\omega \equiv 1/\eta^*$
  - Small country: $\eta^* = \infty \Rightarrow t^* = 0$
  - This is the object Broda, Limão and Weinstein (2008) estimate

Optimal Tariff and Pass-Through

- **Representation 2**: divide by p instead of p^* and use $\frac{dp^*}{dt} = \frac{dp}{dt} - 1$. Let $\rho \equiv \frac{dp}{dt}$ be tariff pass-through to domestic prices and $e_m \equiv -\frac{dm}{dp}\frac{p}{m} > 0$ the import demand elasticity (both evaluated at t^*):

$$\boxed{\frac{t^*}{p} = \frac{1}{e_m} \cdot \frac{1-\rho}{\rho}}$$

- Interpretation
  - Complete pass-through ($\rho = 1$): foreigners bear none of the tariff $\Rightarrow t^* = 0$
  - The lower the pass-through, the larger the share of the tariff paid by foreign exporters, and the larger t^*

- Why this form is useful
  - $\rho < 1$ can come from upward-sloping supply *or* from foreign firms cutting markups
  - Links optimal tariffs to the pass-through literature (e.g. near-complete pass-through of the 2018–19 U.S. tariffs)

Foreign Monopoly: Brander and Spencer (1984)

- A foreign monopolist sells x at home, faces inverse demand $p(x)$, constant marginal cost c, and a specific tariff t; no domestic producer

$$\pi^*(x) = x\big[p(x) - t - c\big] \;\Rightarrow\; \underbrace{p(x) + xp'(x)}_{\text{marginal revenue}} = c + t$$

- Implicit function theorem (SOC: $2p' + xp'' < 0$):

$$\frac{dx}{dt} = \frac{1}{2p'(x) + xp''(x)} < 0, \qquad \rho = \frac{dp}{dt} = \frac{p'(x)}{2p'(x) + xp''(x)}$$

- Incomplete pass-through ($\rho < 1$) $\iff$ $p' + xp'' < 0$ $\iff$ marginal revenue is steeper than demand

Pass-Through and Demand Curvature

- Write the condition in terms of the convexity of demand $\varrho(x) \equiv -xp''(x)/p'(x)$ (Mrázová and Neary):

$$\rho = \frac{1}{2 - \varrho(x)}, \qquad \rho < 1 \iff \varrho(x) < 1$$

- Examples
  - Linear demand: $\varrho = 0$, so $\rho = 1/2$; half of the tariff is paid by the foreign firm
  - CES demand with elasticity ε: $\varrho = (\varepsilon + 1)/\varepsilon > 1$, so $\rho = \frac{\varepsilon}{\varepsilon - 1} > 1$ (over-shifting): a specific tariff *worsens* the TOT

- Lesson: whether a tariff on a foreign monopolist improves the TOT is a question about demand curvature, i.e. about pass-through, which can be measured

Foreign Monopoly: Ad Valorem Tariff

- With an ad valorem tariff τ, the foreign producer price is $p^* = p/\tau$ and the monopolist sets a markup over c:

$$p^* = \frac{\varepsilon(x)}{\varepsilon(x) - 1}\, c, \qquad \varepsilon(x) = \text{home demand elasticity}$$

- Differentiating (Feenstra, ch. 7), with $\varepsilon_x \equiv d\varepsilon/dx$:

$$\frac{dp^*}{d\tau} = -\frac{\varepsilon_x}{(\varepsilon - 1)^2}\, \frac{dx}{d\tau}\, c$$

- Since $dx/d\tau < 0$: TOT gain ($dp^*/d\tau < 0$) $\iff$ $\varepsilon_x < 0$ $\iff$ $\boxed{\varrho(x) < \dfrac{\varepsilon(x) + 1}{\varepsilon(x)}}$, i.e. subconvex demand
  - Weaker than $\varrho < 1$; CES is the knife-edge with constant markups and $dp^*/d\tau = 0$
  - Intuition: lower sales make demand more elastic (Marshall's second law), so the foreign monopolist cuts its markup

Cournot Duopoly: Setup

- So far there is no domestic profit; it matters once a domestic firm competes with the importer
- Inverse demand $p(z)$ with total sales $z = x + y$: x by the foreign firm, y by the home firm. First-order conditions:

$$p(z) + x\,p'(z) = c_x + t \quad \Rightarrow \text{ foreign best response } x = r^*(y, t)$$
$$p(z) + y\,p'(z) = c_y \quad \Rightarrow \text{ home best response } y = r(x)$$

- Best responses slope downward (strategic substitutes); we assume the standard stability condition
- The tariff shifts the foreign best response inward: the foreign firm sells less, and the home firm expands along its best response
- Home iso-profit curves: home profit is higher at lower x, so the tariff raises home profits as foreign sales fall

Cournot Duopoly: Pass-Through

- Add the two FOCs and differentiate with respect to t:

$$\frac{dz}{dt} = \frac{1}{3p'(z) + z\,p''(z)} \;\Rightarrow\; \rho = \frac{dp}{dt} = \frac{p'(z)}{3p'(z) + z\,p''(z)}$$

- $\rho < 1 \iff \boxed{2p'(z) + z\,p''(z) < 0}$: marginal revenue is downward sloping
  - Much weaker than with a foreign monopoly; satisfied by CES ($\varepsilon > 1$) and linear demand
  - So a TOT gain $dp^*/dt < 0$ is generic under Cournot

Cournot Duopoly: Is This Profit Shifting?

- The third term now also helps: $dy/dt > 0$ and $p > C'(y)$, so expanding home output reduces the monopoly distortion

- Feenstra's caveat: do not call this "profit shifting"
  - What matters for welfare is $(p - C')\, dy/dt$, not the change in home profits
  - Under Bertrand competition (or other conduct) home profits can rise while $dy/dt < 0$, which worsens the distortion and offsets the TOT gain
  - Policy prescriptions from strategic trade models are fragile to conduct (Eaton and Grossman, 1986)

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Grossman and Helpman (1994): Environment

- Core idea: the government trades off aggregate welfare against campaign contributions from organized lobbies
- Goods $i = 1, \ldots, N$ plus numeraire; population L, each person supplies one unit of labor (wage 1)

$$V(p, I^h) = I^h + \underbrace{\sum_{i=1}^{N} u_i\big(d_i(p_i)\big) - \sum_{i=1}^{N} p_i d_i(p_i)}_{\text{consumer surplus } S(p)}$$

- Sector-specific capital K_i, owned by H_i people; $H = \sum_i H_i < L$
- Small open economy: world prices p_i^* fixed $\Rightarrow$ no TOT motive
- Profits $\pi_i(p_i) = \max_{L_i} \; p_i f_i(L_i, K_i) - L_i$, with $\pi_i'(p_i) = y_i(p_i)$ (Hotelling)
- Imports $m_i(p_i) = L\, d_i(p_i) - y_i(p_i)$; tariff revenue $T(p) = \sum_i (p_i - p_i^*)\, m_i(p_i)$, rebated per capita

Lobbies and the Government

- Joint welfare of the owners of capital i, and of everyone else:

$$W_i(p) = \pi_i(p_i) + H_i\big[1 + S(p)\big] + \tfrac{H_i}{L}\,T(p), \qquad W_0(p) = (L - H)\big[1 + S(p)\big] + \tfrac{L-H}{L}\,T(p)$$

- Industries split into organized J_o and unorganized J_u. Menu auction (Bernheim and Whinston, 1986): with *truthful* contribution schedules, lobby $j \in J_o$ offers

$$R_j(p) = \max\{0,\ W_j(p) - B_j\}$$

- Government maximizes contributions plus weighted welfare ($\alpha > 0$):

$$\begin{aligned} G(p) &= \sum_{j \in J_o} R_j(p) + \alpha\Big[\sum_{i=1}^{N} W_i(p) + W_0(p)\Big] \\ &= \sum_{j \in J_o} \big[(1+\alpha)W_j(p) - B_j\big] + \sum_{j \in J_u} \alpha W_j(p) + \alpha W_0(p) \end{aligned}$$

- Organized groups get weight $1 + \alpha$; everyone else gets α

Protection for Sale: Equilibrium Tariffs

- Choosing t_j is equivalent to choosing $p_j = p_j^* + t_j$. Let $\delta_j = 1$ if $j \in J_o$ and

$$\lambda_0 \equiv \sum_{i \in J_o} \frac{H_i}{L} \quad \text{(share of the population that owns organized capital)}$$

- First-order condition:

$$\frac{\partial G}{\partial p_j} = (\delta_j + \alpha)\, y_j - (\alpha + \lambda_0)\, L\, d_j + (\alpha + \lambda_0)\left[m_j + t_j \frac{dm_j}{dp_j}\right] = 0$$

- Substitute $m_j = Ld_j - y_j$: $(\delta_j - \lambda_0)\, y_j + (\alpha + \lambda_0)\, t_j\, m_j' = 0$. With $e_j \equiv -m_j'\, p_j / m_j > 0$:

Protection for Sale

$$\frac{t_j}{p_j} = \frac{\delta_j - \lambda_0}{\alpha + \lambda_0} \cdot \frac{y_j / m_j}{e_j}$$

Protection for Sale: Predictions

- Organized sectors ($\delta_j = 1$) get positive protection; unorganized sectors ($\delta_j = 0$) get import *subsidies*, since organized owners also consume these goods
- Protection is higher when
  - Domestic output is large relative to imports (y_j/m_j high): more rent per unit of deadweight loss
  - Import demand is inelastic (e_j low): the deadweight cost of protection is small
  - The government cares little about welfare (α low)
- Limits
  - If everyone were organized ($\lambda_0 = 1$) or nobody ($\lambda_0 = 0$, $\delta_j = 0$), free trade
  - Which industries organize is exogenous here (Mitra, 1999, endogenizes it)
  - Small country: no TOT motive. Grossman and Helpman (1995), "Trade Wars and Trade Talks", add large countries and the TOT term

Testing Protection for Sale: Goldberg and Maggi (1999)

- Multiply the equilibrium condition by e_j and add an error:

$$e_j \frac{t_j}{p_j} = \underbrace{-\frac{\lambda_0}{\alpha + \lambda_0}}_{\gamma} \frac{y_j}{m_j} + \underbrace{\frac{1}{\alpha + \lambda_0}}_{\beta} \delta_j \frac{y_j}{m_j} + u_j$$

- Predictions: $\gamma < 0$, $\beta > 0$, $\gamma + \beta > 0$

- Data: U.S. manufacturing (3-digit SIC), 1983
  - LHS: non-tariff barrier coverage ratios (tariffs are bound by GATT); import elasticities from Shiells, Stern and Deardorff
  - δ_j: organized if PAC contributions exceed a threshold
  - y_j/m_j and δ_j are endogenous: instrumented with industry characteristics

Goldberg and Maggi (1999): Findings

- Signs as predicted: protection rises with y_j/m_j in organized industries and falls in unorganized ones
- Structural parameters: the implied α is large, so the government weighs aggregate welfare far more than contributions (on the order of 50 times or more), and λ_0 is close to one
- Interpretation: the *pattern* of protection fits the model, but the implied departure from welfare maximization is small, consistent with low observed U.S. protection
- See also Gawande and Bandyopadhyay (2000); Mitra, Thomakos and Ulubasoglu (2002) for Turkey

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Why Elasticities?

- Both motives above hinge on elasticities
  - TOT: the optimal tariff equals the inverse export supply elasticity ω
  - Protection for Sale: import demand elasticity e_j
  - Welfare and gains from variety: the elasticity of substitution σ
- Recovering them from trade data needs a model of demand *and* supply
- Workhorse: Feenstra (1994), applied at scale by Broda and Weinstein (2006)
  - Nested CES: imports are a CES aggregate (elasticity γ) of goods g; each good is a CES aggregate (elasticity σ_g) of country-specific varieties i
  - Estimates and code: `http://www.columbia.edu/~dew35/TradeElasticities/TradeElasticities.html`

Demand and Supply of Varieties

- Demand for variety i of good g (share s_{git}, taste shocks ϕ_{gt}, d_{git}):

$$\Delta \ln s_{git} = (\sigma_g - 1)\,\Delta \ln \phi_{gt} - (\sigma_g - 1)\,\Delta \ln p_{git} + \Delta \ln d_{git}$$

- Standard approach: constant markups, horizontal supply, so OLS works if supply and demand shocks are orthogonal
- BW allow upward-sloping supply $p_{git} = \nu_{git}\, q_{git}^{\omega_g}$, $\omega_g \geq 0$, with quantity $q_{git} = s_{git} E_{gt} / p_{git}$ (E_{gt} = total expenditure on g):

$$\Delta \ln p_{git} = \frac{\omega_g}{1+\omega_g}\Big(\Delta \ln s_{git} + \Delta \ln E_{gt}\Big) + \frac{1}{1+\omega_g}\,\Delta \ln \nu_{git}$$

- Assume σ_g, ω_g, and ϕ_{gt} common across exporters; difference relative to a reference country k to remove ϕ_{gt} and E_{gt}:

$$\Delta^k \ln s_{git} = -(\sigma_g - 1)\,\Delta^k \ln p_{git} + \varepsilon^k_{git}, \qquad \Delta^k \ln p_{git} = \frac{\omega_g}{1+\omega_g}\,\Delta^k \ln s_{git} + \delta^k_{git}$$

Identification: Feenstra (1994)

- No natural supply shifters $\Rightarrow$ identify from **second moments** (Leamer, 1981)
  - Assume $E[\varepsilon^k_{git}\,\delta^k_{git}] = 0$ (a strong assumption)
  - Multiply the two equations: for each (g, i),

$$\underbrace{(\Delta^k \ln p_{git})^2}_{Y_{git}} = \theta_1 \underbrace{(\Delta^k \ln s_{git})^2}_{X_{1,git}} + \theta_2 \underbrace{\Delta^k \ln p_{git}\,\Delta^k \ln s_{git}}_{X_{2,git}} + u_{git}$$

with $\theta_1 = \frac{\rho_g}{(\sigma_g-1)^2(1-\rho_g)}$, $\theta_2 = \frac{2\rho_g-1}{(\sigma_g-1)(1-\rho_g)}$, $\rho_g \equiv \frac{\omega_g(\sigma_g-1)}{1+\omega_g\sigma_g}$

- Estimate by averaging over t within each exporter (exporter dummies as instruments), then map $(\theta_1, \theta_2) \mapsto (\sigma_g, \omega_g)$
- What identifies? Heteroskedasticity across exporters: demand-shock variances are similar but supply-shock variances differ, so each exporter traces out a different combination of the two curves (needs ≥ 3 exporters)
- BW (2006): median σ_g of roughly 2.5–4 depending on aggregation; large dispersion across goods; a grid search handles inadmissible estimates

Broda, Limão and Weinstein (2008): Question

- Do countries use tariffs to exploit market power, as the textbook TOT formula $t^* = \omega$ predicts?
- Sample: 15 countries *before* WTO accession
  - Members negotiate tariffs cooperatively, so their tariffs need not reflect ω (Bagwell and Staiger)
  - Non-members set tariffs unilaterally: the right place to test the noncooperative formula
- Method: the Feenstra–BW system, estimated separately for each importer i and HS4 good g, with the object of interest now ω_{ig} rather than σ_g
  - Reference variety differs by importer–good pair: k_{ig}
  - Controls for Protection for Sale in an extension

Broda, Limão and Weinstein (2008): Sample

- Countries and periods before WTO accession

TABLE 1—DATA SOURCES AND YEARS

	GATT/WTO	Production data		Tariff data[a]	Trade data[b]
	Accession date	Source	Years		
Algeria				93	93–03
Belarus				97	98–03
Bolivia[c]	8-Sep-1990	UNIDO	93	93	93–03
China	11-Dec-2001	UNIDO	93	93	93–03
Czech[d]	15-Apr-1993			92	93–03
Ecuador	21-Jan-1996	UNIDO	93	93	94–03
Latvia	10-Feb-1999	UNIDO	96	97	94–03
Lebanon				00	97–02
Lithuania	31-May-2001	UNIDO	97	97	94–03
Oman	9-Nov-2000			92	94–03
Paraguay	6-Jan-1994			91	94–03
Russia				94	96–03
Saudi Arabia	11-Dec-2005			91	93–03
Taiwan	1-Jan-2002	UNIDO	96	96	92–96
Ukraine		UNIDO	97	97	96–02

Can Small Countries Have Market Power?

- Yes, in specialized varieties: an importer can account for a large share of a particular exporter's sales (e.g. "eels from Thailand")

TABLE 2—TRADE AND TARIFF DATA SUMMARY STATISTICS

	Trade data			Tariff data				
				Rate per four-digit HS				
	Number of varieties[a]	Number of HS4 goods	Median # of var. per HS4	Observations[b]	Mean	Standard Deviation	Median	Fraction of HS6 variation between HS4
Algeria	26,466	1,100	13	739	23.8	17.4	15.6	0.95
Belarus	24,440	1,172	12	703	12.4	7.8	10.0	0.94
Bolivia	18,592	1,064	9	647	9.8	0.8	10.0	0.63
China	63,764	1,217	33	1,125	37.9	26.0	30.3	0.93
Czech Republic	61,781	1,219	30	1,075	9.5	17.6	5.1	0.87
Ecuador	22,979	1,101	11	753	9.8	5.5	10.6	0.91
Latvia	33,790	1,128	17	872	7.3	10.5	1.0	0.90
Lebanon	34,187	1,109	15	782	17.1	14.8	15.0	0.87
Lithuania	34,825	1,159	17	811	3.6	7.4	0.0	0.90
Oman	20,482	1,107	10	629	5.7	8.7	5.0	0.76
Paraguay	15,430	1,049	7	511	16.1	11.3	14.0	0.91
Russian	66,731	1,187	34	1,029	10.7	11.0	5.7	0.95
Saudi Arabia	62,525	1,202	32	1,036	12.1	2.6	12.0	0.93
Taiwan	38,397	1,215	19	891	9.7	8.5	7.5	0.90
Ukraine	37,693	1,128	18	730	7.4	7.6	5.0	0.95
Median	34,187	1,128	17	782	9.8	8.7	10.0	0.91

[a] Varieties are defined as six-digit HS, exporting country pairs.
[b] Number of observations for which elasticities and tariffs are available.

Inverse Export Supply Elasticities

- Median ω in the middle tercile ranges from 0.9 to 3.0 across countries (overall median 1.6); the distribution is highly skewed

TABLE 3A—INVERSE EXPORT SUPPLY ELASTICITY STATISTICS

Statistic	Observations[a]	Median[b]			Mean		Standard deviation	
Sample	All	Low	Medium	High	All	W/out top decile	All	W/out top decile
Algeria	739	0.4	2.8	91	118	23	333	47
Belarus	703	0.3	1.5	61	85	15	257	36
Bolivia	647	0.3	2.0	91	102	23	283	49
China	1,125	0.4	2.1	80	92	17	267	35
Czech Republic	1,075	0.3	1.4	26	63	7	233	18
Ecuador	753	0.3	1.5	56	76	13	243	30
Latvia	872	0.2	1.1	9	52	3	239	8
Lebanon	782	0.1	0.9	31	56	7	215	18
Lithuania	811	0.3	1.2	24	65	6	235	16
Oman	629	0.3	1.2	25	209	7	3,536	21
Paraguay	511	0.4	3.0	153	132	67	315	169
Russia	1,029	0.5	1.8	33	48	8	198	18
Saudi Arabia	1,036	0.4	1.7	50	71	11	232	25
Taiwan	891	0.1	1.4	131	90	20	241	43
Ukraine	730	0.4	2.1	78	86	16	254	34
Median	782	0.3	1.6	54	85	13	243	30

[a] Number of observations for which elasticities and tariffs are available. The tariff availability did not bind except for Ukraine, where it was not available for about 130 HS4 goods for which elasticities were computed.

[b] The median over the "low" sample corresponds to the median over the bottom tercile of inverse elasticities. Medium and high correspond to the second and third terciles.

Market Power and Product Differentiation

- Inverse supply elasticities are higher for differentiated goods than for homogeneous goods (Rauch, 1999, classification)

TABLE 5—INVERSE ELASTICITIES BY PRODUCT TYPE

	Differentiated	Reference priced	Commodity
Median inv elasticity	2.38	0.70	0.45
Standard errors	(0.04)	(0.06)	(0.14)
p-value: Differentiated vs. refer. or commod.		0.00	0.00
Mean inv. elasticity	17.5	9.3	8.3
Standard errors	(0.71)	(0.70)	(1.23)
p-value: Differentiated vs. refer. or commod.		0.00	0.00

Notes: The number of observations for the median regression is 8,734, less than the full sample since not all HS4 can be uniquely matched to Rauch's classification. The number for the mean regression is 7,927 because we trim the top decile. The pattern of results with the top decile is similar but with higher values.

Tariffs and Market Power: Within-Country Evidence

- Regress tariffs on market power across goods within countries: $\tau_{ig} = \beta f(\omega_{ig}) + \eta_{iG} + x_{ig}\gamma + u_{ig}$
  - G: broad industry; f: level, log, or a dummy for the top two terciles
  - IV: average ω of the same good in other countries (measurement error; tariffs can feed back into ω_{ig})
- Goods with mid/high market power carry tariffs about 9 percentage points higher, about the size of the average tariff

Tariffs and Market Power: IV Estimates

TABLE 8—TARIFFS AND MARKET POWER ACROSS GOODS (WITHIN COUNTRIES): IV ESTIMATES

Dependent variable	Average tariff at four-digit HS (%)								
Fixed effects	Country			Country and industry			Industry by country		
Estimation method	IV GMM (1)	IV GMM (2)	IV GMM (3)	IV GMM (4)	IV GMM (5)	IV GMM (6)	IV GMM (7)	IV GMM (8)	IV GMM (9)
Inverse exp. elast.	0.040			0.089			0.075		
	(0.027)			(0.055)			(0.028)		
Mid and high inv. exp. elast.		3.96			8.88			9.07	
		(0.76)			(1.18)			(1.08)	
Log(1/export elasticity)			0.75			1.71			1.73
			(0.15)			(0.23)			(0.21)
Observations	12,258	12,258	12,258	12,258	12,258	12,258	12,258	12,258	12,258
No. of parameters	16	16	16	35	35	35	284	282	283
1st stage F	5	1649	1335	2	653	517	3	691	544

Notes: Standard errors in parentheses (heteroskedasticity robust). Industry dummies defined by section according to the Harmonized Standard tariff schedule.

Broda, Limão and Weinstein (2008): Takeaways

- Non-WTO members set systematically higher tariffs on goods supplied inelastically
- Market power explains more of the variation in tariffs than a standard political economy variable
- U.S. non-tariff barriers, which are not covered by the WTO, are also higher where the U.S. has more market power
- Caveats
  - ω_{ig} is estimated with substantial noise, hence the IV and tercile approaches
  - Tariff *levels* are far below ω in the high tercile: market power matters at the margin, but the strict formula $t^* = \omega$ fails
  - Follow-ups: Soderbery (2018) refines the estimator; Nicita, Olarreaga and Silva (2018) show WTO members with "water" in their bindings also price their market power

Outline

1 Welfare Effects of a Tariff

2 Optimal Tariffs

3 Political Economy: Protection for Sale

4 Measuring Market Power in Trade

5 **Nash Tariffs and Trade Agreements**

6 Quantifying Trade Wars and Trade Talks: Ossa (2014)

Two-Country Model

- Home exports good 1 and imports good 2; Foreign the reverse. Ad valorem tariffs τ (Home) and τ^* (Foreign)
- Relative prices: world $p^w \equiv p_2^w / p_1^w$; Home local $p = \tau p^w$; Foreign local $p^* = p^w / \tau^*$ (here * denotes Foreign, not the world price)
- Market clearing $m(p) = x^*(p^*)$ pins down $p^w(\tau, \tau^*)$; absent Metzler and Lerner paradoxes,

$$\frac{\partial p^w}{\partial \tau} < 0 < \frac{\partial p}{\partial \tau}, \qquad \frac{\partial p^*}{\partial \tau^*} < 0 < \frac{\partial p^w}{\partial \tau^*}$$

- Bagwell and Staiger (1999, 2002): any government objective (welfare, Protection for Sale, …) can be written as

$$G\big(p,\ p^w\big), \qquad \text{since tariff revenue } T = (\tau - 1)\, p^w \big[d(p) - y(p)\big]$$

  - Local price p captures domestic (incl. political) concerns; p^w captures the pure TOT effect, with $G_{p^w} < 0$ for an importer of good 2

Nash Tariffs and the Political Optimum

- Nash equilibrium: each government sets its tariff taking the other's as given, with $\frac{\partial p}{\partial \tau} = p^w + \tau \frac{\partial p^w}{\partial \tau} > 0$ the total effect on the local price

$$G_p \frac{\partial p}{\partial \tau} + G_{p^w} \frac{\partial p^w}{\partial \tau} = 0, \qquad G^*_{p^*} \frac{\partial p^*}{\partial \tau^*} + G^*_{p^w} \frac{\partial p^w}{\partial \tau^*} = 0$$

- **Political optimum**: tariffs set as if governments ignored the TOT term, $G_p = 0$ and $G^*_{p^*} = 0$

- Key results
  - Nash tariffs are inefficient, and the *only* source of inefficiency is the TOT externality, even with arbitrary political objectives
  - Politically optimal tariffs are efficient: they lie on the contract curve
  - Each government shifts part of the cost of its tariff onto the other, so Nash tariffs are too high: a trade war

What Is the GATT/WTO For?

- If the TOT externality is the only problem, an agreement should neutralize it and leave domestic (incl. political) objectives alone
- Implications for the GATT/WTO
  - **Reciprocity**: matched tariff cuts leave p^w unchanged, which neutralizes the TOT motive
  - **MFN**: a single world price per good, so reciprocity works multilaterally
- Grossman and Helpman (1995): the same logic with lobbies; “trade wars” versus “trade talks”

Outline

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6 Quantifying Trade Wars and Trade Talks: Ossa (2014)

Ossa (2014): Motivation

- First comprehensive quantitative analysis of *noncooperative* and *cooperative* trade policy
  - How large are optimal tariffs? What tariffs would a world trade war produce?
  - How costly would a breakdown of cooperation be? How much is left to negotiate?

- One framework that nests all three motives from this lecture
  - TOT (Johnson, 1953–54), via relative wages
  - "Profit shifting" (Venables, 1987), via cross-industry markup differences
  - Political economy (Grossman and Helpman, 1994), via industry weights

- Why only now? Calibrated GE trade models (post-EK 2002), exact hat algebra (DEK 2007), and MPEC optimization (Su and Judd, 2012; Knitro) made industry-level tariff games solvable

- Predecessors: Perroni and Whalley (2000, Armington, implausibly high Nash tariffs), Ossa (2011, Krugman), BLW (2008, statistical)

Headline Answers

	Without lobbying	With lobbying
Average optimal tariff (no retaliation)	$\approx$ 59%	62%
Gain of tariff-imposing country	2.2%	1.9%
Loss of other countries	-0.7%	-0.7%
Average Nash (trade war) tariff	58%	63%
Welfare change	-2.4%	-2.9%
Gains from trade talks starting at: Nash tariffs	3.4%	3.6%
factual tariffs	0.5%	0.5%
zero tariffs	≈ 0	0.2%

Welfare in percent of real income (government welfare G with lobbying). Tariffs are averages of country medians. Source: Ossa (2014), text and Tables 3–7.

Setup: Multi-Sector Krugman–Ricardo

- N countries, S industries. Love of variety within industries, Cobb–Douglas across:

$$U_j = \prod_s \left(\sum_i \int_0^{M_{is}} x_{ijs}(\nu_{is})^{\frac{\sigma_s - 1}{\sigma_s}} d\nu_{is} \right)^{\frac{\sigma_s}{\sigma_s - 1} \mu_{js}}$$

- Labor only, iceberg costs θ_{ijs}, productivity φ_{is}: $\quad l_{is} = \sum_j \theta_{ijs} x_{ijs} / \varphi_{is}$
- Constant markups: $p_{is} = \frac{\sigma_s}{\sigma_s - 1} \frac{w_i}{\varphi_{is}}$; number of firms M_{is} **fixed** (no free entry), so industries earn profits
- Tariffs t_{ijs}, $\tau_{ijs} \equiv 1 + t_{ijs}$ (imports into j from i); no other instruments
- Why fixed M_{is}? Positive profits give lobbies something to fight over, and no corner solutions with zero production

Setup: Politically Motivated Governments

- Government objective with political economy weights λ_{js} (mean one across s):

$$G_j = \sum_s \lambda_{js} W_{js}, \qquad W_{js} = \frac{w_j L_{js} + \pi_{js} + \frac{L_{js}}{L_j} TR_j}{P_j}$$

  - $\lambda_{js} = 1$ for all s: $G_j = W_j$, real income
  - Industry revenue is split in fixed shares between workers ($\frac{\sigma_s - 1}{\sigma_s}$) and profits ($\frac{1}{\sigma_s}$), so both lobby for higher industry sales

- Interpretation: a dollar of income in industry s counts λ_{js} times as much as a dollar in an industry with average political support
- Reduced form of Protection for Sale, but includes workers as well as capital owners
- Tariff revenue TR_j is rebated to industries in proportion to employment

Equilibrium Conditions in Levels

For given tariffs, an equilibrium is $\{w_i, X_i, P_{js}, \pi_{is}\}$ such that

$$\pi_{is} = \frac{1}{\sigma_s} \sum_j M_{is}\, \tau_{ijs}^{-\sigma_s} \left(\frac{\sigma_s}{\sigma_s - 1} \frac{\theta_{ijs} w_i}{\varphi_{is} P_{js}} \right)^{1-\sigma_s} \mu_{js} X_j \qquad \text{(profits)}$$

$$w_i L_i = \sum_s \pi_{is}(\sigma_s - 1) \qquad \text{(labor market)}$$

$$P_{js} = \left(\sum_i M_{is} \left(\frac{\sigma_s}{\sigma_s - 1} \frac{w_i \theta_{ijs} \tau_{ijs}}{\varphi_{is}} \right)^{1-\sigma_s} \right)^{\frac{1}{1-\sigma_s}} \qquad \text{(price index)}$$

$$X_j = w_j L_j + \sum_i \sum_s t_{ijs} M_{is}\, \tau_{ijs}^{-\sigma_s} \left(\frac{\sigma_s}{\sigma_s - 1} \frac{\theta_{ijs} w_i}{\varphi_{is} P_{js}} \right)^{1-\sigma_s} \mu_{js} X_j + \sum_s \pi_{js} \qquad \text{(budget)}$$

- $2N(S+1)$ equations in $2N(S+1)$ unknowns, given a numeraire
- Problem: depends on $\{M_{is}, \theta_{ijs}, \varphi_{is}\}$, which are hard to estimate

Equilibrium Conditions in Changes (Exact Hat Algebra)

With $\widehat{x} = x'/x$ and T_{ijs} = flows at world prices, an equilibrium is $\{\widehat{w}_i, \widehat{X}_i, \widehat{P}_{js}, \widehat{\pi}_{is}\}$ such that

$$\widehat{\pi}_{is} = \sum_j \alpha_{ijs}\, \widehat{\tau}_{ijs}^{-\sigma_s}\, \widehat{w}_i^{1-\sigma_s}\, \widehat{P}_{js}^{\sigma_s - 1}\, \widehat{X}_j, \qquad \alpha_{ijs} \equiv \frac{T_{ijs}}{\sum_n T_{ins}}$$

$$\widehat{w}_i = \sum_s \gamma_{is}\, \widehat{\pi}_{is}, \qquad \gamma_{is} \equiv \frac{\frac{\sigma_s - 1}{\sigma_s} \sum_j T_{ijs}}{\sum_t \frac{\sigma_t - 1}{\sigma_t} \sum_n T_{int}}$$

$$\widehat{P}_{js} = \left(\sum_i \beta_{ijs} \left(\widehat{w}_i\, \widehat{\tau}_{ijs} \right)^{1-\sigma_s} \right)^{\frac{1}{1-\sigma_s}}, \qquad \beta_{ijs} \equiv \frac{\tau_{ijs} T_{ijs}}{\sum_m \tau_{mjs} T_{mjs}}$$

$$\widehat{X}_j = \frac{w_j L_j}{X_j}\, \widehat{w}_j + \sum_i \sum_s \frac{t'_{ijs} T_{ijs}}{X_j}\, \widehat{w}_i^{1-\sigma_s} \widehat{P}_{js}^{\sigma_s - 1} \widehat{\tau}_{ijs}^{-\sigma_s} \widehat{X}_j + \sum_s \frac{\pi_{js}}{X_j} \widehat{\pi}_{js}$$

- Needs only σ_s, tariffs, and trade flows: $X_j = \sum_i \sum_s \tau_{ijs} T_{ijs}$, $\pi_{is} = \frac{1}{\sigma_s} \sum_j T_{ijs}$, $w_j L_j = X_j - \sum_i \sum_s t_{ijs} T_{ijs} - \sum_s \pi_{js}$
- Implicitly calibrates $\{M_{is}, \theta_{ijs}, \varphi_{is}\}$ to match T_{ijs} exactly; reduces to $2N$ equations in $\{\widehat{w}_i, \widehat{X}_i\}$

Data and Calibration

- 7 regions (Brazil, China, EU, India, Japan, U.S., Rest of World) $\times$ 33 agricultural and manufacturing industries, 2007; trade, production and applied tariffs from GTAP 8
- **Trade imbalances**: constant nominal deficits behave badly under large tariffs (and cannot hold as tariffs $\to \infty$)
  - First purge the data of deficits DEK-style (set transfers to zero in the model), then run all counterfactuals on the balanced data
- **Elasticities** σ_s: Feenstra (1994) on UN Comtrade 1994–2008, pooled across the main importers
  - Mean 3.42; homogeneous goods highest (wheat 10.1, rice 7.0), other animal products lowest (1.9)
- **Political economy weights** λ_{js}: calibrated later from noncooperative tariffs
- Benchmark: autarky costs 10–21% of real income (U.S. 13.5%), more than one-sector models because σ_s differ (Ossa, 2015)

Parameter Estimates

	σ_s	λ_{BRA}	λ_{CHN}	λ_{EU}	λ_{IND}	λ_{JPN}	λ_{ROW}	λ_{US}
Wheat	10.07	1.29	1.68	1.66	1.67	1.91	1.27	1.24
Rice	7.01	1.23	1.64	1.51	1.38	1.91	1.23	1.19
Dairy	5.89	1.24	1.29	1.56	1.29	1.46	1.25	1.34
Wearing apparel	5.39	1.29	1.46	1.22	1.13	1.03	1.22	1.48
Other metals	4.47	1.11	0.80	1.06	1.05	0.94	1.08	1.16
Vegetable oils, etc.	4.03	1.13	1.19	1.09	1.50	0.96	1.08	1.03
Bovine meat products	3.89	1.05	1.16	1.44	1.05	0.95	1.11	1.07
Leather products	3.67	1.18	1.13	1.06	0.97	1.17	1.08	1.22
Ferrous metals	3.67	1.10	0.73	1.04	1.04	0.90	1.03	1.09
Other manufactures	3.53	1.15	1.22	1.06	1.00	0.86	1.04	1.23
Other cereal grains	3.32	0.98	1.22	1.26	1.04	1.72	1.41	0.91
Oil seeds	3.21	0.97	1.01	0.87	1.20	1.79	1.20	1.04
Other meat products	3.20	1.03	1.09	1.12	1.04	0.91	1.11	0.96
Beverages and tobacco products	2.92	1.01	1.45	1.28	1.51	0.93	1.09	1.24
Bovine cattle, etc.	2.91	0.92	0.56	1.07	0.92	0.94	0.91	0.81
Textiles	2.87	1.04	1.19	0.96	0.91	0.82	0.99	1.26
Other transport equipment	2.84	0.91	0.76	0.94	0.74	0.73	0.90	1.03

Notes: The entries under "σ_s" are the estimated elasticities of substitution. The entries under "λ_{BRA}" until "λ_{US}" are the estimated political economy weights for each country which are scaled to have a mean of one. The boxes highlight the industries with the five highest values of λ for each country.

Parameter Estimates (cont.)

	σ_s	λ_{BRA}	λ_{CHN}	λ_{EU}	λ_{IND}	λ_{JPN}	λ_{ROW}	λ_{US}
Plant-based fibers	2.80	0.96	0.58	0.77	0.82	0.73	0.90	0.81
Wool, etc.	2.76	0.91	0.85	0.00	0.86	1.37	0.91	0.64
Motor vehicles, etc.	2.75	0.99	1.18	0.99	0.96	0.73	0.92	0.90
Metal products	2.70	1.01	0.96	0.94	0.84	0.75	0.91	1.06
Sugar	2.69	0.92	1.22	1.10	1.34	1.27	1.00	0.85
Other food products	2.62	0.90	1.01	0.98	1.05	0.93	0.96	0.89
Paper products, etc.	2.56	0.93	0.67	0.88	0.81	0.73	0.86	0.91
Other crops	2.53	0.92	0.64	0.87	1.12	0.74	0.93	0.80
Electronic equipment	2.49	0.91	0.82	0.87	0.61	0.66	0.84	1.00
Other mineral products	2.47	0.92	0.87	0.87	0.80	0.72	0.89	1.02
Other machinery, etc.	2.46	0.94	0.76	0.86	0.79	0.73	0.83	1.00
Vegetables, etc.	2.42	0.82	0.96	0.83	1.01	0.90	0.90	0.80
Chemical products, etc.	2.34	0.85	0.74	0.84	0.73	0.71	0.81	0.96
Wood products	2.32	0.90	0.77	0.79	0.72	0.71	0.83	0.90
Forestry	2.20	0.77	0.78	0.66	0.61	0.70	0.76	0.61
Other animal products	1.91	0.68	0.59	0.56	0.50	0.72	0.73	0.53
Mean	3.42	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Notes: The entries under "σ_s" are the estimated elasticities of substitution. The entries under "λ_{BRA}" until "λ_{US}" are the estimated political economy weights for each country which are scaled to have a mean of one. The boxes highlight the industries with the five highest values of λ for each country.

General Equilibrium Effects of a Tariff

- Example: the U.S. raises its tariff by 50 percentage points in one industry

	Chemicals ($\sigma = 2.34$)	Apparel ($\sigma = 5.39$)
Δ U.S. wage	$+1.45\%$	$+0.67\%$
Δ U.S. output, protected industry	$+5.73\%$	$+33.35\%$
Δ U.S. output, other industries (avg.)	-1.40%	-0.97%
Δ U.S. welfare	$+0.17\%$	-0.14%

- Mechanism: expenditure shifts to U.S. goods $\Rightarrow$ protected industry expands $\Rightarrow$ bids up U.S. wages $\Rightarrow$ other industries contract. Mirror image abroad
- TOT effect is positive in both cases (higher relative wage). Profit shifting differs: positive for the high-markup industry (chemicals), negative for the low-markup one (apparel)

Welfare Decomposition

- Log-linearizing around factual tariffs (Ossa, 2014, eq. 14):

$$\frac{\Delta W_j}{W_j} \approx \underbrace{\sum_i \sum_s \frac{T_{ijs}}{X_j}\left(\frac{\Delta p_{js}}{p_{js}} - \frac{\Delta p_{is}}{p_{is}}\right)}_{\text{terms of trade}} + \underbrace{\sum_s \frac{\pi_{js}}{X_j}\left(\frac{\Delta \pi_{js}}{\pi_{js}} - \frac{\Delta p_{js}}{p_{js}}\right)}_{\text{profit shifting}} + \underbrace{\sum_i \sum_s \frac{t_{ijs} T_{ijs}}{X_j}\left(\frac{\Delta T_{ijs}}{T_{ijs}} - \frac{\Delta p_{is}}{p_{is}}\right)}_{\text{trade volume}}$$

- Each term mirrors the Feenstra formula from the first part of the lecture: TOT, production efficiency (now through markups $\sigma_s/(\sigma_s - 1)$), and deadweight loss

Interpreting the Decomposition

- **TOT**: since $p_{is} \propto w_i$, TOT $=$ relative wages; a tariff in one industry moves the TOT in *all* industries (this form of the term uses balanced trade, $\sum_{i,s} T_{jis} = \sum_{i,s} T_{ijs}$)
- **Profit shifting**: real profit changes from changes in industry *output*, weighted by profit shares $\propto 1/\sigma_s$
  - With a common σ the term is zero: the expansion of one industry is exactly offset by the contraction of others
  - With free entry it becomes Venables' production relocation effect (extensive margin)
- **Trade volume**: tariff revenue from changes in import quantities (Section 1's deadweight term)
- First-order only; Flam–Helpman markup-correction motives enter at second order

Computing Optimal, Nash and Cooperative Tariffs

- **Optimal tariffs** (others' tariffs at factual levels): MPEC (Su and Judd, 2012)

$$\max_{\{\tau_{ijs}\}_{i\neq j,\, s}} \widehat{G}_j \quad \text{s.t. equilibrium conditions in changes}$$

  - Treat $\{\widehat{w}, \widehat{X}\}$ as choice variables and the equilibrium conditions as constraints; avoids nested fixed point
  - $(N-1) \times S = 198$ tariffs per country; solved with Knitro

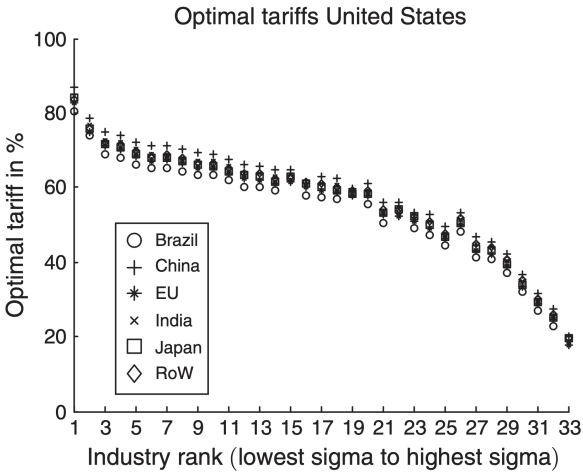
- **Nash tariffs**: iterate best responses (each country's optimal tariffs given the others') until convergence; robust to starting values (autarky is always also an equilibrium)

- **Cooperative tariffs**: symmetric Nash bargaining, all governments gain equally

$$\max_{\{\tau\}} \widehat{G}_1 \quad \text{s.t.} \quad \widehat{G}_j = \widehat{G}_1 \;\; \forall j, \;\; \text{equilibrium conditions}$$

  - Threat points: Nash tariffs (as in theory), factual tariffs (scope for future rounds), zero tariffs (distance of free trade from the frontier)

Optimal Tariffs without Lobbying



- U.S. panel shown (markers = trading partners); medians 54–62% across countries
- Strongly *decreasing* in σ_s: low-σ industries have high markups (profit shifting) and give more market power
- Highly correlated across countries (mean correlation 0.91)
- Larger countries set higher tariffs and impose larger externalities
- Common $\sigma = 3.42$: uniform 44% tariffs (pure TOT)
- Trade of the imposing country falls 31%

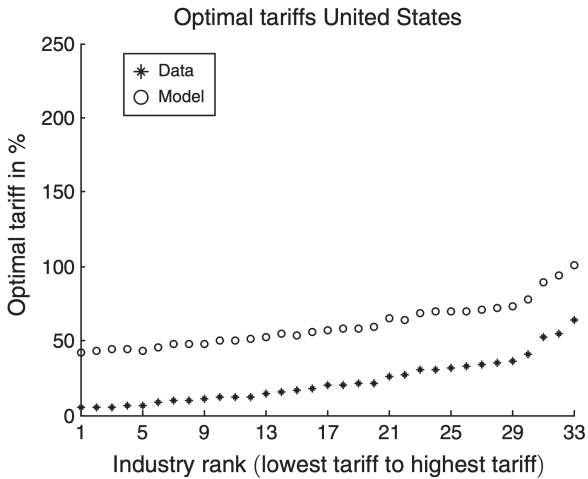
Calibrating the Political Economy Weights

- Factual tariffs are the outcome of negotiations and their link to optimal tariffs is unclear; their cross-industry ranking is even negatively correlated with optimal tariffs (-0.16)

- Instead, match the *cross-industry distribution* of optimal tariffs to observed **noncooperative** tariffs, after removing means (least squares, $\lambda_{js} \geq 0$)
  - U.S. "column 2" tariffs (Cuba, North Korea): average 23% vs. 3% applied
  - China and Japan "general rate" tariffs; EU "autonomous rate" tariffs
  - Brazil, India, Rest of World: factual tariffs, since large "water" between bound and applied rates
  - Optimal tariffs restricted to be nondiscriminatory for this exercise

- Results (Table 1): most favored U.S. industries are apparel, dairy, textiles, beverages and tobacco, wheat
  - Weights correlated across countries (rank correlation 0.70) and with σ_s (0.78): politically motivated protection is concentrated where the efficiency motive calls for low tariffs

Optimal Tariffs with Lobbying



- U.S. panel: stars are column 2 tariffs, circles model optimal tariffs, industries ranked by data tariff
- Model matches the cross-industry shape but overpredicts levels for all countries except China
- Average of median optimal tariffs: 62.4%
- Aggregate effects similar to no lobbying; profit shifting all but vanishes as political weights drive the cross-industry pattern
- Japan's 721% rice tariff is capped in the matching

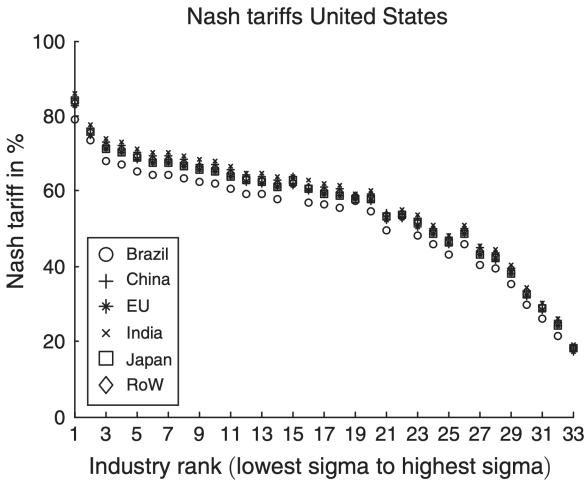
Sensitivity to the Trade Elasticity

- Scale all σ_s so that the mean matches the range of trade elasticities in Simonovska and Waugh (2014)

Mean σ	Average of median optimal tariffs	
	without lobbying	with lobbying
3.5	56.8%	60.1%
5.0	34.3%	35.5%
6.5	24.6%	25.6%

- Optimal tariffs, and the gains from imposing them, fall steeply with σ: lower elasticities mean more market power
- With mean $\sigma = 6.5$, U.S. optimal tariffs nearly match U.S. column 2 tariffs
- Rankings of political weights are stable across scalings (rank correlation 0.97)
- Lesson: the trade elasticity is again the key parameter (compare the Gravity lecture)

Trade Wars: Nash Tariffs



- Nash tariffs look like optimal tariffs: median 58.1%, close to the ≈50% average tariff after Smoot–Hawley (1930)
- Nobody wins: welfare falls by 2.4% on average (2.9% with lobbying, tariffs 63.4%)
- Trade flows fall by 58%
- Losses largest for the most open region (Rest of World, import share 27%); Japan loses least because its factual farm tariffs exceed its Nash tariffs
- Common σ: uniform 43% tariffs, -1.8%

Trade Talks: Cooperative Tariffs

- Structure of efficient tariffs
  - Cross-industry: tariffs now *increase* in σ_s, taxing low-markup goods to offset markup differences across industries (Flam and Helpman, 1987), since domestic instruments are unavailable
  - Cross-country: tariffs and import subsidies move relative wages to replicate side payments so that all gain equally (Mayer, 1981)
  - Ruling out import subsidies changes little
- Gains from negotiations (without lobbying)
  - From Nash: $+3.4$% for every country, an upper bound on the value of the WTO system
  - From factual tariffs: $+0.5$%; about 85% of the potential gains have already been realized
  - From free trade: negligible, so free trade is close to the efficiency frontier; tariffs are a poor tool for domestic distortions (targeting principle)
- With lobbying: 3.6%, 0.5%, 0.2%; similar conclusions

Ossa (2014): Assessment

- Contributions
  - Brings Bagwell–Staiger and Grossman–Helpman to the data in a single GE model
  - Template for later work: exact hat algebra + MPEC to compute optimal and equilibrium policies

- Limitations, and where the literature went
  - One factor and constant markups: TOT = relative wages, so tariffs cannot target industry-specific TOT
  - No intermediate inputs or input–output linkages: Caliendo and Parro (2015); Bagwell, Staiger and Yurukoglu (2021) model actual GATT bargaining with MFN
  - Fixed entry: free entry and scale economies change the industrial policy motive (Lashkaripour and Lugovskyy, 2023)
  - Balanced trade: Caliendo, Kortum and Parro (2025) study tariffs with deficits
  - Sufficient-statistics versions of the tariff war: Lashkaripour (2021); survey in Ossa (2016)
- Current interest: carbon border adjustments and climate clubs (Farrokhi and Lashkaripour, 2025; Hsiao, 2025)

Summary

- One formula, three motives: $\frac{dW}{dt}$ = deadweight loss + TOT + production efficiency
- Optimal tariff: $t^*/p^* = 1/\eta^*$ (inverse export supply elasticity), equivalently $t^*/p = \frac{1}{e_m}\frac{1-\rho}{\rho}$: pass-through is the empirical counterpart
- Imperfect competition: demand curvature determines whether a tariff improves the TOT
- Protection for Sale explains the cross-industry pattern of protection; welfare weight is large
- Non-WTO countries price their market power (BLW, 2008)
- TOT externalities make Nash tariffs inefficient, which rationalizes reciprocity and MFN (Bagwell and Staiger)
- Quantitatively: optimal and Nash tariffs around 60%, a trade war costs about 3% of real income, and most gains from cooperation have been realized (Ossa, 2014)

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